Magnetic separation (8 points)

Part A. Magnetic susceptibility of materials

A.1	1.1pt

$\langle p_z \rangle =$ _____

Answer: _____

A.2	1.1pt

$\alpha \to 0$: $\langle p_z \rangle =$ _____

$\alpha \to \infty$: $\langle p_z \rangle =$ _____

(sketch on the working sheets)

Answer: _____

A.3	0.9pt

$\chi =$ _____

Answer: _____

Part B. The magnetic separator

B.1	0.8pt

$B_0 =$ _____

$a =$ _____

Answer: _____

B.2	0.8pt

$E =$ _____

Answer: _____

B.3	1.1pt

$x =$ _____

Answer: _____

B.4	1.1pt

$\Delta_0 =$ _____

Answer: _____

B.5	1.1pt

$\omega_{\min} =$ _____

$\omega_{\max} =$ _____

Answer: _____